

TRIAL RISK • EARLY-WARNING DASHBOARD

Catching termination risk early — across three tiers

A hybrid of M365 Copilot and on-device local LLMs. The AI detects, aggregates, and triages — it never makes the safety or termination call. A walkthrough.

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Illustrative throughout. Participant-level data runs on-device, zero egress; validated tools own every reported number.



PARTICIPANT

DLTs • safety • PK — on-device



STUDY

stopping rules • enrollment • QTLs



CLIENT

commercial early-warning

Termination risk hides until the meeting — we want it sooner

Early-phase trials can stop for reasons that build quietly: a participant's safety signal, a cohort's toxicity trend against the stopping rule, or a sponsor cooling on the program. Too often the first formal look is the cohort review or the governance meeting — days or weeks late. The dashboard surfaces all three, continuously.

PARTICIPANT

A dose-limiting toxicity emerging in the DLT window — a Hy's-Law liver signal, QTc prolongation, an AE cluster escalating.

STUDY

The cohort DLT rate approaching the stopping rule; enrollment or quality limits breached; a futility signal.

CLIENT / SPONSOR

Payment aging, communication cooling, scope-reduction change-orders, sponsor pipeline or funding stress.

Three RAG panels and a live early-warning feed



Legend: ● termination-level ● watch / precursor ● within limits

Illustrative current state. Each panel rolls weighted signals up to a RAG status — not an autonomous verdict.

Participant-level safety — processed on-device, zero egress

The AI flags candidate DLTs and safety patterns early, before the formal cohort review, and pre-assembles the SRC packet. It never adjudicates the DLT.

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|--|--|
| ● Candidate DLTs in the evaluation window | by design — 3+3 / BOIN / mTPI-2 / CRM; sentinel dosing |
| ● Hy's-Law / drug-induced liver injury (eDISH) | ALT/AST >3×ULN + bili >2×ULN + ALP <2×ULN |
| ● QTcF prolongation | >500 ms, or Δ >60 ms from baseline |
| ● PK exposure outliers | C _{max} / AUC beyond the NOAEL-scaled safety margin |
| ● AE-cluster & grade escalation; SAE / death | CTCAE grade trend; expedited-reporting clock |

Engine: on-device local LLM. Decision: the medical monitor / Safety Review Committee.

Dose-escalation rules, enrollment, quality — aggregate KRIs

De-identified, aggregate signals run on cloud Copilot; any participant-level drill-down stays on-device. These watch the study's trajectory against its own rules.

- **Cohort DLT rate vs the stopping / de-escalation rule** excess toxicity → escalation stop / MTD reached early
- **Enrollment & screen-failure KRIs** vs the recruitment plan; feasibility
- **Safety Quality Tolerance Limits (ICH E6(R3))** aggregate SAE / AE-rate QTL breaches
- **Data-quality & central-monitoring signals** query aging, protocol-deviation trend, site anomalies
- **Futility & probability of meeting the objective** interim signals against the plan

Engine: cloud Copilot (aggregate) + on-device drill-down. Decision: the PM, with the SRC / DSMB on safety.

TIER 3 • CLIENT / SPONSOR

Commercial early-warning — decision-support for the account team

Non-sensitive business data on cloud Copilot. This tier is signal and triage for the account team — never a verdict the system renders about a sponsor.

- **Accounts-receivable / payment aging**

invoice disputes; days-past-due breaches

- **Communication cadence & sentiment**

meeting / email drop; relationship cooling

- **Scope-reduction & change-order trend**

de-scope often precedes cancellation

- **Sponsor financial & pipeline health**

runway, layoffs, portfolio re-prioritization, M&A

- **Competitive readouts & contract / renewal window**

adjacent-program failures; option-exercise signals

Engine: cloud Copilot (CRM / finance / public filings). Decision: the account / BD lead.

Sensitivity decides the engine; humans decide the outcome

ON-DEVICE LOCAL LLM

participant-level safety / PK / unblinded

- **Zero egress** — data never leaves the validated environment.
- Carries the whole participant tier and any participant-level drill-down.
- Part-11 audited.

M365 COPILOT (CLOUD)

non-sensitive ops & commercial

- **De-identified, aggregate study KRIs and the client / commercial tier.**
- Under BAA; no PHI or unblinded data.
- Where it can't reach, the local model does.

The rule that governs everything

The AI flags, aggregates, triages, and drafts the alert — it never makes the safety or termination call. The medical monitor / SRC / DSMB make every DLT and safety determination; the PM owns the study; the account lead owns the client signal. Reported safety numbers come from the validated safety database, the cohort DLT-rate from the validated dose-escalation tool — never the LLM. Every flag triggers a human review and a logged disposition; nothing is paused or terminated by the system itself.

A continuous, anticipatory layer on top of the rules

Optional, and decision-support only — it surfaces risk earlier and continuously, but the threshold rules stay authoritative and it is validated before any regulated use.

OPTIMAL TRANSPORT

Wasserstein drift

How far has a distribution drifted from a meta-learned reference? Wasserstein distance respects clinical severity — it sees mass move toward the dangerous tail where a single threshold is silent — and points to which analyte is driving it.

ACTIVE INFERENCE

free energy + EFE

Treat the monitor as a model of the expected trajectory. Rising surprise (free energy) is an early warning; Expected Free Energy picks the most informative next assessment — say, an unscheduled liver panel — and recommends it for approval.

NESTED + META

one deep hierarchy

Participant, study, and client are one model: priors flow down, risk flows up, and the priors are meta-learned across studies. A participant signal propagates to study and program risk — not three siloed dashboards.

What it buys over thresholds alone: earlier · geometry- & tail-aware · anticipatory · coherent across tiers. Methods independently math-checked.

A DLT early-warning fires — day 9, not day 21

Day 9, cohort 4

Participant S-1042's central-lab feed posts ALT 3.4×ULN, total bilirubin 2.1×ULN, ALP 1.3×ULN. The on-device model evaluates the eDISH logic, recognizes a Hy's-Law pattern, and — inside the DLT window — flags a candidate DLT. Nothing left the device.

Within minutes

It pre-assembles the case: the LFT trajectory, the eDISH context, dose & exposure, concomitant meds, and the running cohort DLT tally against the 3+3 rule. It drafts the alert and routes it to the medical monitor and SRC chair — ahead of the scheduled review — with a Part-11 audit entry.

The humans decide

The medical monitor orders a confirmatory repeat panel; the SRC convenes. Per protocol, further dosing pauses pending adjudication. The validated safety database holds the authoritative numbers; the SRC makes the DLT call.

The payoff

The signal surfaced on day 9, not at the cohort meeting on day 21 — SRC packet already built, data never leaving the device, and a human making every call.

Safe wins first, then the on-device tier, then scoring & audit

1

The safe wins — cloud

- Wire the Study & Client tiers: enrollment / KRI / QTL feeds and the finance-AR / CRM / sponsor-comms signals into cloud Copilot.
- Build the Study and Client RAG panels behind a human-review queue. De-identified / non-sensitive only.

2

The on-device participant tier

- Stand up the local LLM in the validated environment; connect safety DB / central-lab / ECG; encode the DLT, Hy's-Law/eDISH, QTcF and PK-exposure logic.
- Route every participant-level signal on-device, zero egress; wire alerting to the medical monitor / SRC and the disposition log.

3

Scoring, escalation & audit

- The weighted roll-up to tier RAG, the escalation routing, the Part-11 audit trail and human-acknowledgement, the cross-study safety view.
- Optionally layer in the advanced engine — validated and frozen before it informs any regulated decision.

Catch termination risk before the meeting — and let humans make every call.

Earlier

Drift and surprise rise before a line is crossed; a Hy's-Law signal on day 9, not day 21.

Three tiers, one screen

Participant safety, study stopping-rules, and client/commercial — the earliest mover is visible.

Governed by design

On-device for sensitive data; validated tools own the numbers; the SRC / DSMB / PM decide.

Fits the program

A Phase-1 green-pilot deliverable on the same hybrid stack — no new principle, new target.